

Customer Churn Prediction with KNN

Part 2: Unguided Exercise - sklearn Model Training + Evaluation

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1. Approach

The goal was to predict whether a telecommunications customer would churn using a K-Nearest Neighbors (KNN) classifier. The workflow followed the lab requirements: data exploration, preprocessing, an 80/20 stratified train/test split, KNN training, evaluation, comparison of multiple k values, and business interpretation.

Preprocessing included converting TotalCharges to numeric, replacing its 11 blank values with the median, removing customerID, mapping Churn to 0/1, and one-hot encoding categorical variables. The numeric features SeniorCitizen, tenure, MonthlyCharges, and TotalCharges were standardized after the split because KNN relies on distances and is sensitive to feature scale.

2. Model Performance

The required k values (1, 3, 5, 7, 9, 11, 15) were compared using accuracy, precision, recall, and F1 score. **k = 9** produced the best overall balance on the test set.

k	Accuracy	Precision	Recall	F1
1	73.4%	49.9%	51.3%	50.6%
3	75.4%	53.5%	55.3%	54.4%
5	76.4%	55.3%	57.5%	56.4%
7	76.4%	55.4%	57.2%	56.3%
9	78.0%	58.6%	58.6%	58.6%
11	77.8%	58.2%	57.8%	58.0%
15	77.8%	58.4%	56.7%	57.5%

Selected model (k = 9): accuracy 78.0%, precision 58.6%, recall 58.6%, F1 58.6%. Confusion matrix: 880 true negatives, 155 false positives, 155 false negatives, and 219 true positives.

Because only 26.5% of customers churn, a model that always predicts “No Churn” would already achieve about 73.5% accuracy. The KNN model improves accuracy by about 4.5 percentage points, but recall shows that it still misses roughly 41% of actual churners. Therefore, accuracy alone is not sufficient for judging business usefulness.

3. Key Findings About Churn

Exploratory analysis showed the strongest churn associations in the following groups:

- Month-to-month contracts had a churn rate of about 42.7%, compared with 11.3% for one-year and 2.8% for two-year contracts.
- Fiber-optic customers churned at about 41.9%, compared with 19.0% for DSL customers.
- Electronic-check customers had a churn rate of about 45.3%, substantially above automatic payment methods.
- Churners had much shorter average tenure (about 18 months vs. 38 months) and higher average monthly charges (about 74.4 vs. 61.3).
- Customers without online security or technical support also showed noticeably higher churn rates.

These are associations, not proof of causation. KNN itself does not provide native feature-importance scores.

4. Business Recommendations

- Prioritize retention outreach to newer customers on month-to-month contracts, especially fiber-optic subscribers.
- Investigate service experience and pricing for fiber-optic customers and consider targeted retention or support offers.
- Review the high churn among electronic-check customers and test incentives for automatic payment methods.
- Use the model as a screening tool for retention teams rather than an automatic decision system; in practice, optimize for churn recall and expected retention value/cost.

5. Limitations and Future Improvements

- The target is imbalanced and the selected model still misses 155 of 374 churners in the test set.
- Results come from one train/test split; cross-validation would provide a more robust estimate.
- KNN can become inefficient with many one-hot encoded features and offers limited interpretability.
- Compare KNN with Logistic Regression or Random Forest, and add PR/ROC metrics, threshold tuning, and business-cost analysis.